EVERY MODULE HAS A FREE RESOLUTION, AND A FINITE MODULE OVER A NOETHERIAN RING HAS A FINITE FREE ONE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Every module is a quotient of a free module. Used at the first step of the counted proof (line 17647). Starts the resolution with a surjection from a free module onto M.
- Finitely generated submodules of finite free modules over a Noetherian ring are finite. Used at the inductive step of the counted proof (lines 17648-17651). Makes the kernel of each map in the resolution finitely generated, so the next free module can be taken finite.

Lemma 0.1. *Let R be a ring. Let M be an R-module.*

(1) *There exists an exact complex*

$$\ldots \to F_2 \to F_1 \to F_0 \to M \to 0.$$

with F_i free R-modules.

(2) *If R is Noetherian and M finite over R, then we can choose the complex such that F_i is finite free. In other words, we can find an exact complex*

$$\ldots \to R^{\oplus n_2} \to R^{\oplus n_1} \to R^{\oplus n_0} \to M \to 0.$$

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Proof. Let us explain only the Noetherian case. As a first step choose a surjection $R^{n_0} \to M$. Then having constructed an exact complex of length e we simply choose a surjection $R^{n_{e+1}} \to \mathrm{Ker}(R^{n_e} \to R^{n_{e-1}})$ which is possible because R is Noetherian. $\square$